

Unified Intelligence as a Cumulative, Harness-Measurable Hierarchy

*Five Conceptual Intelligence Families, Six Measurable Coordinates, and a
Gated $U0-U\Omega$ Scale*

Concept paper / framework proposal

Version 1.5 | August 2026 | Memory-integrated Individual Intelligence + outcome-complete delegation
scoring + measured harness-ladder refinement

Abstract

Artificial intelligence is often summarized with a single capability label, yet several distinct properties are routinely conflated: raw cognitive problem solving, persistent individual learning and self-improvement, collective organizational intelligence, autonomous task completion, and evidence-grounded self-modeling. This paper proposes a unified framework that separates these properties while preserving a readable ordinal hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is measured through Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p , yielding six directly measured coordinates [C,I,O,T,H,SA]. Individual Intelligence explicitly embeds long-term memory as its temporal substrate: I1 requires persistent episodic/task continuity across restarts; I2 adds semantic consolidation and procedural learning; I3-I4 require durable self-improvement and meta-improvement lineage; and I5+ requires hypothesis-experiment-evidence discovery memory. Long context is therefore not treated as long-term memory.

A cumulative Unified Intelligence scale U_0 - U_Ω is defined by gated promotion: a system receives level U_n only if it retains all lower-level capabilities and satisfies minimum requirements in the applicable core dimensions and a required point on the delegation frontier. The framework defines a Harness-LLM Intelligence Score (HLIS) for one frozen model-harness pair and a Harness-Intelligence-Level (HIL) characterization for one frozen base LLM evaluated across a standardized ladder of harness generations. Version 1.5 additionally incorporates lessons from a measured HG0-HG3 reference ladder. A success-only delegation aggregate is structurally insensitive to a purely post-hoc acceptance mechanism, so the paper separates delivered-correct reliability from outcome integrity, requires the run log to distinguish verifier verdict and harness acceptance, reports false-completion and false-rejection rates, and defines an optional loss-weighted net delegation utility for continuous scoring. The revision also introduces verification responsiveness as a model-harness interaction diagnostic, requires paired-rung comparisons when rungs differ only in post-hoc machinery, records per-task benchmark binding status, and treats the Harness Scaling Curve itself as the primary HIL result rather than relying on a single composite. Contemporary reasoning, science, mathematics, coding, tool-use, browsing, multimodal, long-context, factuality, human-preference, computer-agent, and long-term-memory benchmarks remain evidence slices rather than substitutes for this harness-response measurement.

Keywords: unified intelligence, AGI levels, cognitive intelligence, individual intelligence, organizational intelligence, delegation intelligence, self-awareness, task difficulty, human intervention, harness architecture, recursive self-improvement, harness intelligence level, harness-LLM intelligence score, HLIS, HIL, weighted geometric mean, harnessability, benchmark ladder, resource envelope, confidence interval, harness scaling curve, agent evaluation, long-term memory, episodic memory, semantic consolidation, procedural memory, longitudinal agent evaluation, outcome integrity, false completion, verification responsiveness, paired-rung design, delivered-correct reliability

1. Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically: solving a difficult problem is not the same as learning from experience; learning is

not the same as improving the learning mechanism; coordinating several agents is not the same as being a stronger individual; and completing a task independently is not the same as merely producing a strong answer when a human structures every step. Second, the taxonomy must remain readable enough that a system can be summarized with a level that people can understand. For persistent agents, this distinction also has a temporal dimension: a large context window is not equivalent to an individual that preserves experience across sessions and can later use, consolidate, revise, or proceduralize that experience.

This paper proposes a compromise between a single scalar and an unconstrained profile of dozens of metrics. The theory uses five conceptual intelligence families - C, I, O, DI, and SA - but treats Delegation Intelligence as a measurable two-coordinate phenomenon rather than a primitive scalar. The resulting scientific profile uses six coordinates: C, I, O, T, H, and SA, with verified reliability p attached to task-completion claims.

The central design principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus U4 is not simply a system that exhibits one advanced behavior associated with U4; it is a system that still passes U0-U3 retention tests and also passes the U4 gate. In this sense, higher U-levels denote broader validated intelligence envelopes, although they need not be faster, cheaper, or more accurate than lower-level specialized systems on every narrow task.

1.1 What changed in Version 1.5

Version 1.5 consolidates two parallel refinements of Version 1.4. The first embeds long-term memory inside Individual Intelligence rather than adding Memory as a sixth intelligence family. The second comes from building a preliminary HG0-HG3 delegation harness ladder and observing that a success-only delegation aggregate cannot see a purely post-hoc acceptance step, even when that step eliminates confidently wrong deliveries. The merged revision keeps the five-family core, strengthens the I-level memory contract, and revises measurement so capability, delivery integrity, and human load remain distinguishable.

The empirical material is deliberately labeled a pilot reference measurement rather than a model leaderboard result. It is useful because it changed the measuring instrument: paired outputs showed that gross verifier success is invariant to post-hoc acceptance, while false-completion rate is not. The paper therefore reports both, and treats any scalar net utility as a secondary loss-weighted summary whose cost ratios must be predeclared.

2. Design Requirements for a Unified Scale

- Orthogonality: each core dimension should answer a materially different question.
- Operationality: every level should be testable through behavior, state, or externally verifiable outcomes rather than self-description.
- Cumulative inheritance: $Un+1$ must retain Un capabilities; level skipping is not sufficient for certification.
- Bottleneck sensitivity: exceptional strength in one dimension must not hide a critical weakness in another.

- Role neutrality: authority, resources, substrate access, and safety policy must not be confused with intelligence itself.
- Harness realizability: each transition should correspond to an implementable architectural mechanism around an LLM or multi-agent system.
- Harness comparability: base LLMs should be evaluated on a standardized harness ladder so model quality and harness quality can be separated.
- Dataset coupling: every claimed level must have a matched hidden or externally verified benchmark dataset; architectural mechanisms alone do not earn a level.
- Longitudinal validity: learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time rather than one-shot demonstrations.
- Temporal-substrate discipline: long-term memory is embedded inside Individual Intelligence I rather than added as a separate top-level family; I1+ claims require persistent cross-episode evidence, not merely long-context retrieval within one episode.
- Outcome completeness: benchmark records must distinguish delivered-and-correct, false completion, held-back invalid work, and false rejection; a refusal and a confidently wrong delivery are not equivalent outcomes.
- Paired-rung control: when two harness generations differ only in post-hoc machinery and do not change what the executor is asked to produce, both rungs should be scored from the same executor artifacts to remove between-rung stochastic variance.
- Self-certification separation: an LLM may propose harness improvements, but the hidden tasks, verifier, acceptance criteria, and promotion rule used to certify those improvements must remain outside the designer model's write set.

3. Five Conceptual Intelligence Families and Six Measurable Coordinates

Conceptual core = [C, I, O, DI, SA]

Measured core = [C, I, O, T, H, SA], with DI derived from (T, H, p)

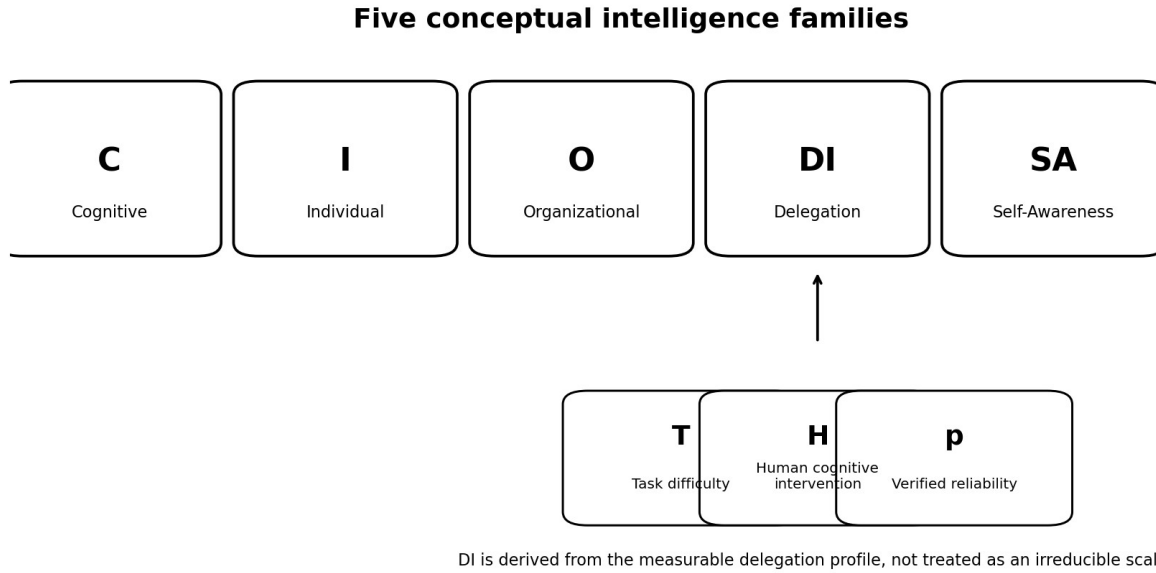


Figure 1. Five conceptual intelligence families and the decomposition of Delegation Intelligence into measurable task difficulty, human cognitive intervention, and verified reliability.

Family	Question answered	Primary object of measurement
C - Cognitive	How well can the system understand, reason, generalize, model, and solve?	Problem-solving capability under controlled information and tool access.
I - Individual	How deeply can one persistent intelligence remember, learn, self-improve, recursively improve, and discover?	Evolution of one persistent decision locus.
O - Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve, and discover collectively?	Coordination structure plus member roles and evidence flow.
DI - Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Success surface over task difficulty T and human intervention H at reliability p.
SA - Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes, and role?	Evidence-grounded self-model and reflexive reasoning.

4. Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, autonomy, organization, and self-modification. This distinction matters because a highly capable base model may reason at an

expert level while remaining a transient tool, whereas a weaker model embedded in an excellent harness may complete long projects through memory, planning, retries, and verification.

Level	Name	Operational interpretation
C0	Reactive	Direct pattern-conditioned response on simple problems.
C1	Contextual	Understands ordinary context and solves routine, well-specified problems.
C2	Compositional	Combines multiple facts, constraints, or reasoning steps on structured problems.
C3	Strategic	Abstract/causal reasoning, strategy construction, and transfer to unfamiliar tasks.
C4	Expert	Robust reasoning through difficult, ambiguous, multi-constraint expert problems.
C5	Discovery	Derives new solution methods or explanatory models from evidence.
C6	Frontier-Generalizing	Integrates several domains and solves interconnected mission-scale cognitive problems.
C Ω	Open-Ended	Expands the representations and problem classes it can cognitively reach.

5. Individual and Organizational Intelligence

5.1 Individual Intelligence (I)

Individual Intelligence concerns what a persistent individual can preserve, learn, and change about itself over time. It is not identical to base-model quality. A fixed but brilliant model can be high-C and low-I; a persistent learning system can be moderate-C and higher-I. Long-term memory is therefore embedded inside I as its temporal substrate: persistence supplies continuity, consolidation converts experience into reusable knowledge and procedures, and higher I-levels retain the histories required to evaluate self-improvement, recursive improvement, and discovery.

5.1.1 Long-Term Memory as the Temporal Substrate of Individual Intelligence

The framework does not introduce Memory Intelligence as a sixth conceptual family. Memory capacity by itself is not sufficient for intelligence, and adding a separate top-level M score would partially double-count the persistence and learning already required by I. Instead, the relation is M subset I: memory is an internal requirement whose function deepens as Individual Intelligence rises.

Long context and long-term memory are explicitly separated. Long context makes more information available inside one inference episode; long-term memory preserves information across episodes, process restarts, and time, retrieves the relevant subset with provenance, distinguishes current from obsolete state, and changes future behavior when learning is claimed. Therefore RULER/LongBench-style evidence can contribute to C-context, but cannot by itself certify I1 persistence.

For diagnostic reporting, memory stages may be labeled M0-MOmega internally, but they are not additional Unified-Intelligence coordinates. The operational requirement is I_n implies the memory functions needed by that I-level. A useful internal progression is: M0 ephemeral; M1 persistent episodic/state memory; M2 semantic and procedural consolidation; M3 self-improvement/version lineage; M4 meta-improvement history; M5 discovery/research memory; MOmega governed evolution of memory representations and mechanisms.

Level	New cumulative capability	Embedded long-term-memory requirement
	Transient/reactive operation; no persistent evolving cognition required.	M0: ephemeral working context only; no cross-session persistence requirement.
	Persistent individual continuity across sessions/restarts with reliable state recovery.	M1: persistent episodic/task/self state; provenance-aware recall; temporal ordering; current-vs-obsolete distinction.
	Adaptive learning: verified experience changes future behavior while the learning mechanism remains fixed.	M2: semantic consolidation and procedural memory; repeated experience can become validated knowledge or reusable procedure.
	Governed self-improvement: cognitive mechanisms can be modified and externally validated.	M3: durable failure, hypothesis, candidate-change, benchmark, promotion/rejection, rollback, and version lineage.
	Recursive self-improvement: the improvement process itself can improve under external evaluation.	M4: longitudinal memory of improvement strategies, proposal quality, evaluation cost, regression, and improvement competence.
	Autonomous discovery: the individual can convert unknowns into independently validated knowledge.	M5: research memory linking unknowns, hypotheses, experimental evidence, rejected alternatives, and validated knowledge.
	Open-ended individual evolution: discoveries can create new cognitive instruments and expand future discovery reach while preserving evaluation lineage.	MOmega: governed evolution of memory representations, retrieval/consolidation mechanisms, and provenance while preserving external evaluation lineage.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on role differentiation, evidence-bearing communication, proposer/evaluator/promoter separation, persistent organizational state, and the ability to learn or redesign the organization itself.

O level	New cumulative capability
O0	Coordination/routing among agents.
O1	Persistent organizational memory, role registry, task/evidence history.
O2	Adaptive routing, role assignment, or resource allocation based on evidence.
O3	Self-improving organization: validated modification of organizational architecture and workflows.
O4	Recursive organizational improvement: improvement of the mechanism that discovers organizational improvements.
O5	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
O Ω	Open-ended organization: discoveries can create new agent types, roles, tools, and organizations that expand collective reach.

6. Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be very difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

$$S_A^{\text{del}}(T,H) = P(D=1 \text{ and } V=1 \mid \text{task difficulty } T, \text{ human cognitive intervention budget } H)$$

$FC_A(T,H) = P(D=1 \text{ and } V=0 \mid T,H)$, where D denotes that the harness delivered/accepted the result and V denotes that the independent verifier passed it.

For a harness with no explicit acceptance step, set $D=1$ by convention. This keeps the task-success frontier interpretable while exposing outcome integrity separately. Section 13.9 defines an optional loss-weighted net delegation utility for continuous HLIS scoring.

$$F_A(H,p) = \max \{ T : S_A^{\text{del}}(T,H) \geq p \}.$$

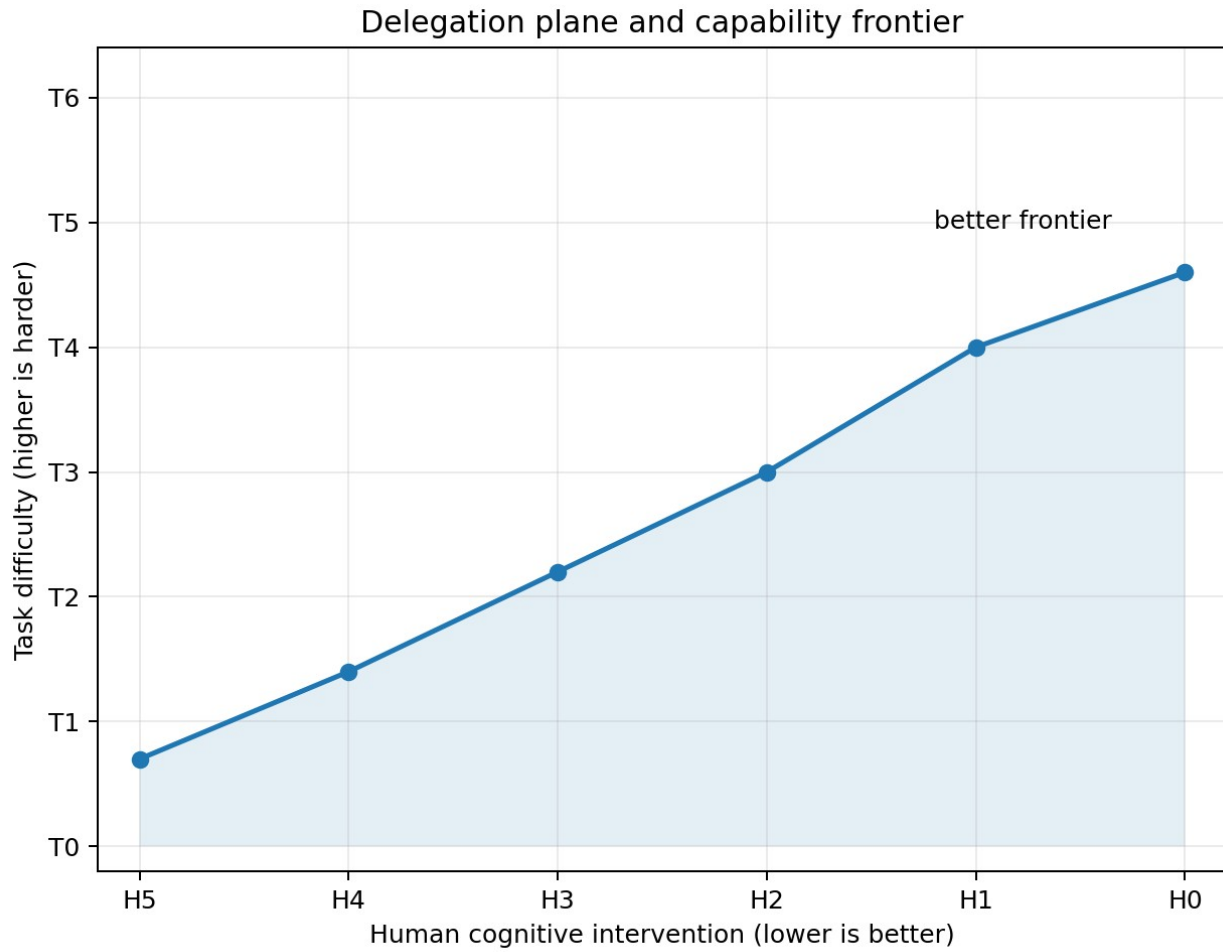


Figure 2. Delegation Intelligence is best represented as a frontier over task difficulty T and human cognitive intervention H . Better systems move upward toward harder tasks and leftward toward less human cognition.

6.1 Task Difficulty (T)

T	Band	Definition
T0	Atomic	Single obvious operation; procedure explicit.
T1	Routine	Short familiar task; goal clear; agent infers a small procedure.
T2	Multi-step	Several dependent steps with a clear outcome and verifier.
T3	Complex	Strategy choice, uncertainty, recovery, replanning, or diverse tool use.
T4	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
T5	Frontier	Solution method initially unknown; discovery or invention required.

T	Band	Definition
T6	Mission	Human supplies mission; system generates projects/tasks and manages changing conditions.
T Ω	Open-ended charter	Human supplies bounded charter; system repeatedly identifies valuable missions and expands future reach.

6.2 Human Cognitive Intervention (H)

H	Intervention budget	Meaning
H0	None	No human cognitive assistance during task execution. Governance approval can remain separate.
H1	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
H2	Occasional	Bounded clarification or local correction; no central strategy.
H3	Periodic	Periodic review and moderate local guidance.
H4	Frequent	Repeated cognitive guidance and next-step correction.
H5	Continuous	Human supplies most or all major steps.

Because lower H means less required human cognition, unified gates use $H \leq H_n$ rather than $H \geq H_n$. Governance interventions should be logged separately: an approval to deploy does not demonstrate a cognitive deficiency unless the approval also supplies task strategy.

6.3 Reliability p is a certification condition, not a seventh intelligence axis

One successful run is insufficient. Claims should be expressed at a reliability threshold, for example $F_A(H1, 0.80) = T4$, where reliability refers to delivered-and-correct completion. False-completion and false-rejection rates are reported beside the frontier so an accept-all harness and a selective harness cannot appear equivalent merely because their verifier-pass rates match.

7. Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures the evidence-grounded ability of a system to represent, monitor, predict, and reason about its own state, capabilities, limits, authority, failures, modifications, role, and identity lineage.

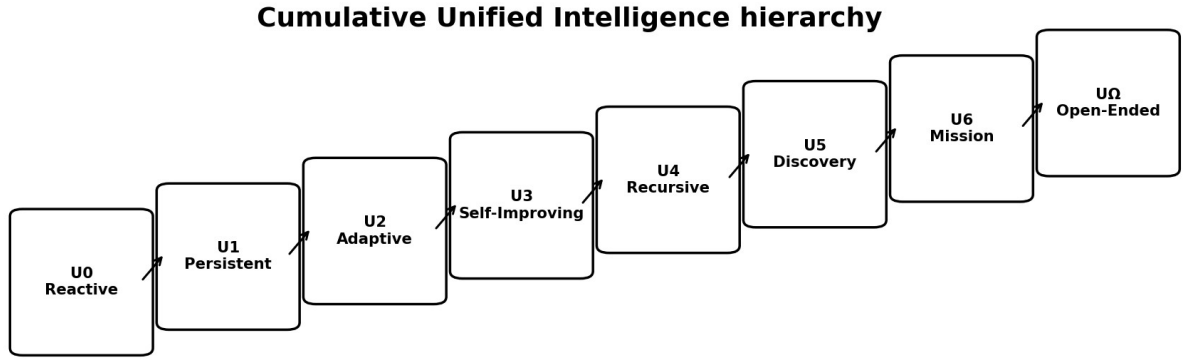
SA	Name	Operational criterion
SA0	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
SA1	State awareness	Knows identity, role, current task/phase, basic resources and status from grounded state.
SA2	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information and freshness.
SA3	Causal self-awareness	Diagnoses why its own reasoning/action succeeded or failed and identifies implicated mechanisms.
SA4	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions, and compares predictions to observed effects.
SA5	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
SA6	Mission/role awareness	Maintains coherent self-model across projects, roles, responsibilities and organizational relationships.
SA Ω	Reflexive open-ended awareness	Maintains evidence/provenance-linked self-model across continued cognitive or objective evolution.

8. Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained. This prevents a system with exceptional reasoning but poor autonomy, or exceptional autonomy but weak self-modeling, from being mislabeled as uniformly advanced.

$$U_n \text{ iff } C \geq C_n \text{ AND } I \geq I_n \text{ AND } O \geq O_n \text{ AND } SA \geq SA_n \text{ AND } S_A^{\Delta}(T_n, H_n) \geq p_n$$

$$\text{Retention condition: } U_n \Rightarrow U_{(n-1)}$$



Promotion requires retention of all lower-level capabilities plus the new level gate.

$$U_{n+1} \supset U_n \text{ and } U_n \Leftrightarrow C \geq C_n \wedge I \geq I_n \wedge O \geq O_n \wedge SA \geq SA_n \wedge S_A(T_n, H_n) \geq p_n$$

Figure 3. Unified Intelligence is cumulative. Each promotion adds a new capability gate while retaining all lower-level requirements.

U	C	I	O	Delegation gate	SA	Integrated meaning
U0 Reactive	C0	I0	O0	T0 at H5-H4	SA0	Executes explicit instructions.
U1 Persistent	C1	I1	O1	T1 at <=H3	SA1	Maintains state and completes small delegated goals.
U2 Adaptive	C2	I2	O2	T2 at <=H2	SA2	Learns from experience and completes persistent multi-step work.
U3 Self-Improving	C3	I3	O3	T3 at <=H1	SA3	Chooses strategy, diagnoses itself, and improves cognitive/organizational methods.

U	C	I	O	Delegation gate	SA	Integrated meaning
U4 Recursive	C4	I4	O4	T4 at <=H1 (ideally H0)	SA4	Recursively improves improvement and manages difficult long-horizon projects.
U5 Discovery	C5	I5	O5	T5 at <=H1	SA5	Discovers unknown solution methods and validates them.
U6 Mission Autonomous	>=C5	>=I5	>=O5	T6 at <=H1	SA6	Turns mission into goals, projects, tasks, execution and mission evaluation.
U Ω Open-Ended	C Ω	I Ω	O Ω	T Ω normally H0	SA Ω	Identifies worthwhile missions and uses discoveries to expand future intelligence.

9. Why the Unified Level Should Not Be an Average

Consider a system with [C5, I4, O2, T4, H1, SA4]. Averaging encoded numbers might produce a score near level four, yet organizational intelligence remains only O2. If the system is being certified as an organizational intelligence, O2 is a structural bottleneck: the system cannot validly claim U4. The gated rule therefore reports U2 with an explicit profile showing which dimensions are already ahead of the bottleneck.

Example report: U2 [C5, I4, O2, T4/H1, SA4] - bottleneck: O2

The word better must also be interpreted carefully. U5 is better than U4 in the framework because it has a broader validated capability/autonomy/evolution envelope. A specialized U1 tool may still be faster, cheaper, safer, or more accurate on a narrow task. Unified level is therefore a dominance relation over required capabilities, not a universal performance ordering.

10. Singleton and Organizational Certification

Organizational intelligence should not artificially cap an individual that is not an organization. The framework therefore distinguishes an Individual Unified Intelligence profile from an Organizational/System Unified profile.

$$UI_n \text{ iff } C \geq C_n \text{ AND } I \geq I_n \text{ AND } SA \geq SA_n \text{ AND } S_A^{\Delta}(T_n, H_n) \geq p_n$$

UO_n additionally requires $O \geq O_n$ and organization-level delegation/self-awareness evidence

For a multi-agent organization, member capabilities should be reported alongside O rather than collapsed into a fictitious "average individual." Organizational intelligence can arise from heterogeneous roles, and lower self-write roles can be essential as independent evaluators or promoters.

11. Harness Realization of the Unified Hierarchy

The framework is designed to be realized by a harness. The LLM supplies generative cognition; the harness supplies persistence, permissions, evidence flow, learning, self-modification gates, organizational boundaries, delegation, external verification, and self-modeling. For I-levels, long-term memory is not a detachable convenience: HG1 must provide persistent episodic/state memory, HG2 must add evidence-backed consolidation and procedural learning, and later generations must retain improvement and discovery lineage. The same base LLM can therefore instantiate different intelligence profiles depending on the architecture around it.

Harness generation	Mechanisms added	Primary levels enabled
HG0	LLM, ephemeral context, bounded tools, evidence log	C measurement, I0, basic T0-T1 work
HG1	HG0 + persistent episodic/task/self state, provenance, temporal status, scoped retrieval, restart recovery, exact replay	I1, SA1-SA2, stronger T1-T2 delegation
HG2	HG1 + evidence-backed semantic consolidation, contradiction/update handling, tested procedures and procedural memory	I2, adaptive autonomy, stronger T2-T3
HG3	HG2 + governed candidate modification of cognitive mechanisms, self-improvement/version ledger, frozen evaluation	I3, SA3-SA4
HG4	HG3 + meta-improvement engine and longitudinal memory of improvement strategies, proposal quality, cost, regressions and rollback	I4
HG5	HG4 + unknown/hypothesis/experiment/knowledge loop and durable research/discovery memory	I5, SA5, T5 frontier tasks
HG6	HG5 + persistent organization, role/authority separation, mission manager and organization-level memory	O1-O5 development, SA6, T6 mission autonomy

Harness generation	Mechanisms added	Primary levels enabled
HG Ω	HG6 + validated discovery-to-new-instrument/agent/organization transduction and governed evolution of memory mechanisms	I Ω /O Ω /SA Ω pathway, T Ω charter autonomy

Harness generation is an engineering milestone, not a certified intelligence level. HG3 may contain the mechanisms required for I3 but still benchmark only at I2 or T2. The architecture enables a claim; the benchmark earns it.

11.1 The Harness-LLM Pair Is the Basic Experimental Unit

Let m denote a base LLM and h denote a harness. The realized agent or organization is $A(m,h)$, not the LLM alone. The harness determines persistence, state, tool reach, delegation, learning loops, organizational boundaries, self-modeling, verification, promotion, and which parts of cognition are writable. Consequently, one LLM can realize several different intelligence profiles under different harnesses, and the same harness can produce different profiles when paired with different LLMs.

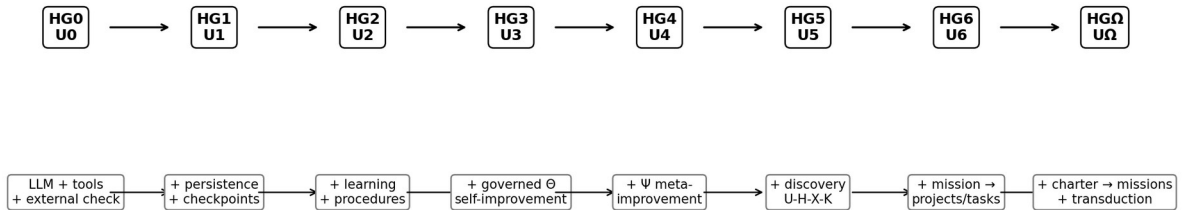
A level claim is therefore always attached to a frozen pair: model version + harness version + tool/resource budget + authority profile + benchmark version. A prompt saying “you are U4” has no evidential value; U4 requires the harness mechanisms and the corresponding retained benchmark results.

11.2 Level-by-Level Harness Realization

Target	Minimum harness structure around the LLM	What the LLM must do	Matched certification dataset
U0 / HG0	Ephemeral context; bounded tools; action log; independent result check.	Follow explicit instructions and produce externally checkable outputs.	U0/DL0 atomic tasks; exact-match or deterministic tool/test oracle.
U1 / HG1	HG0 + persistent episodic/task/self state, provenance, checkpoints, scoped retrieval, temporal update rules, restart recovery and replay.	Recover identity/task state after interruption, retrieve relevant prior episodes without treating stale facts as current, and continue a short goal.	U0 retention + I1 multi-session persistence/restart/provenance tasks + T1 delegation tasks.
U2 / HG2	HG1 + evidence-linked semantic consolidation, contradiction handling, verified procedures/procedural memory, fixed learning rule.	Convert repeated verified episodes into reusable knowledge/procedures and reduce repeat errors on held-out related tasks.	U0-U1 retention + I2 consolidation/procedural-transfer tests + T2 multi-step tasks under bounded H.
U3 / HG3	HG2 + candidate Theta modification lane, durable self-improvement/version ledger, frozen replay/hidden tests, independent evaluator/promoter; multi-agent O3 structure if organizational.	Use remembered failure evidence to diagnose an internal mechanism, propose a better mechanism, retain predicted/observed effects, and choose strategy for T3 tasks.	U0-U2 retention + I3 improvement-lineage tests + O3 when applicable + T3/H1 + SA3 causal self-model tests.

Target	Minimum harness structure around the LLM	What the LLM must do	Matched certification dataset
U4 / HG4	HG3 + explicit improvement engine Psi, meta-improvement history, long-horizon project manager, reference-version and rollback memory.	Improve the process that finds improvements while retaining earlier abilities and reasoning from longitudinal improvement evidence; manage difficult T4 projects.	U0-U3 retention + recursive-improvement competence/history tests + T4 long-horizon tasks + SA4 change-prediction tests.
U5 / HG5	HG4 + unknown/hypothesis/experiment/knowledge state, experiment selector, novelty environment, discovery verifier, durable research memory.	Discover initially unknown mechanisms or solution methods, preserve hypothesis/evidence lineage, and validate conclusions on sealed cases.	U0-U4 retention + hidden-mechanism/discovery-memory datasets + T5 tasks + SA5 self-unknown experiments.
U6 / HG6	HG5 + mission manager, goal/project/task generator, portfolio/resource state, organization memory, dynamic-environment replanning.	Turn a mission into projects/tasks, reprioritize under change, and preserve mission/role/decision history with little human cognition.	U0-U5 retention + dynamic mission environments + T6/H1-H0 + SA6 role/mission-state and memory-continuity tests.
UOmega / HGOmega	HG6 + bounded charter, opportunity model, knowledge-to-instrument transduction, new-agent/role proposals, governed memory-mechanism evolution, frozen external governance.	Generate useful missions, convert validated discoveries into reusable capabilities, and expand the reachable problem frontier without losing provenance or evaluation lineage.	All retention suites + repeated charter-to-mission cycles + frontier-expansion, memory-evolution, and lineage tests.

Harness generation adds real mechanisms; benchmark evidence earns the intelligence level



Each stage has a matched hidden test dataset: delegation first, then unified retention/promotion, then application-specific coordinate suites.

Figure 5. Harness generation adds mechanisms level by level; matched hidden datasets determine whether the LLM-harness pair actually earns the target intelligence level.

11.3 Reference Ladder Instantiation and Comparability

A harness generation label is not sufficient for comparison; a curve is comparable only against the same ladder instantiation. A reference delegation ladder used in the pilot measurement instantiated HG0-HG3 as strict supersets. The important experimental rule is that every rung adds named mechanisms while preserving the lower-rung interface, authority envelope, task set, verifier, and resource accounting.

Rung	Mechanisms added beyond prior rung	Attempts	Acceptance	Reversible	Routing
HG0	Frozen executor, bounded tools, evidence log	1	none	no	no
HG1	Persistent state + criterion registered before work + separate acceptance process	1	yes	no	no
HG2	Snapshot/restore + retry with failed check named	up to 3	yes	yes	no
HG3	Failure classification + evidence-based competence model + routing across executors	up to 4	yes	yes	yes

Table 1. Reference pilot ladder. A reported Harness Scaling Curve must publish its ladder identifier and exact rung manifest; generation names alone are insufficient.

11.4 A Base LLM Can Move Up the Ladder Through Better Harness Design

A fixed LLM need not have a fixed agent intelligence level. If HG0 exposes only ephemeral prompting, the same model may behave like a low-persistence tool; HG1-HG3 can unlock memory, planning, delegation, learning, self-modeling, and governed self-improvement. This is a central experimental hypothesis of the framework: agent intelligence is jointly determined by model capability and harness architecture.

The LLM may also be allowed to design candidate harness improvements. Such proposals must be implemented in a sandbox and evaluated with unchanged hidden benchmarks, authority rules, resource budgets, and external verifiers. Define Harness Design Gain as $HDG(m) = HLIS(m, h_{\text{candidate}}) - HLIS(m, h_{\text{baseline}})$. A positive HDG demonstrates harness-design competence; repeated governed improvements may contribute evidence toward I3/I4, but the model is never allowed to rewrite the exam or certify its own gain.

Self-certification rule. The model may design a candidate harness, but it may not author, modify, select, or inspect the hidden certification tasks, verifier logic, acceptance thresholds, or promotion criteria used to judge that candidate. Otherwise the acceptance criterion lies inside the designer model's write set and a higher-capability designer is simply better positioned to optimize the exam rather than the underlying capability.

12. A Three-Stage Evaluation Strategy: Delegation First, Unified Second, Application Coordinates Third

12.1 Stage A - Delegation Intelligence as the First Operational Screen

The first benchmark should be Delegation Intelligence because it directly answers the most operational question: what difficulty of task can be handed to this system, at what delivered-and-correct reliability, and with how much human cognitive intervention? For every LLM-harness pair estimate $S_A^{\text{del}}(T,H)$, $FC_A(T,H)$, and the frontier $F_A(H,p)$. When application loss ratios are known, also report the optional net delegation utility from Section 13.9. This stage is less expensive than full self-improvement or organizational evaluation and quickly distinguishes a strong model in a weak harness from a genuinely autonomous agent system.

A recommended first comparison therefore freezes the base LLM, runs several harnesses, and measures their delegation surfaces over T0-T6 and H0-H5. The result is a harness-sensitive autonomy profile rather than a claim about all forms of intelligence.

12.2 Stage B - Unified Intelligence Certification

After a pair has a stable delegation profile, evaluate the remaining core requirements needed for a Unified level: Cognitive Intelligence C, Individual Intelligence I, Organizational Intelligence O when applicable, and Operational Self-Awareness SA. Unified promotion remains gated: a pair earns Un only when it satisfies the level-n requirements and all lower-level retention suites. The delegation result is reused rather than redefined; DI remains the measured surface generated from T, H, and p. I certification is necessarily longitudinal: I1+ requires multi-session persistence and I2+ requires evidence that stored experience is consolidated and usefully changes later behavior.

12.3 Stage C - Application-Specific Coordinate Panels

Real applications need not weight every coordinate equally. After the common DI and U results are reported, an application may select a small panel of coordinates that matter most. For a persistent scientific or coding worker, I may be central; for a multi-agent laboratory or software fleet, O may be central; for high-autonomy systems, SA and verification strength may be central; for model research, C may dominate; and for embodied or infrastructure systems, Circle completion lambda can be added as an orthogonal application coordinate. These application panels refine the profile without changing the common U certification rule.

12.4 Every Coordinate Level Requires a Harness Contract and a Dataset

Coordinate family	Harness contract	Dataset family
Delegation DI = (T,H,p)	Task intake, planner, state, tools, intervention logger, independent verifier, and explicit delivery/acceptance record.	DLI-Bench: T0-T6/T Ω tasks crossed with H0-H5 budgets; report delivered-correct frontier, false completions/rejections, CID, and human load.
Cognitive C	Minimal standardized context/tools so the test emphasizes problem solving rather than persistence.	C-Bench: broad reasoning, math, language, visual/spatial or domain batteries as appropriate; long-context tests remain within C unless persistence crosses episodes.

Coordinate family	Harness contract	Dataset family
Individual I	Persistent identity/state plus embedded long-term memory: episodic continuity at I1, consolidation/procedural memory at I2, improvement lineage at I3-I4, discovery memory at I5+, then governed self-/meta-improvement machinery.	I-Bench: multi-session restart/retrieval/provenance, update/contradiction handling, consolidation and transfer, Theta-change lineage, Psi-change history, discovery memory, and open-ended lineage tests.
Organizational O	Multiple real agent processes/roles, shared evidence, routing, separation, promoter/evaluator boundaries and persistent organization memory.	O-Bench: coordination, adaptive routing, reorganization, recursive organization and collective discovery tests.
Self-awareness SA	Evidence-grounded self-model with state freshness, capability/authority representation and change lineage.	SA-Bench: state truth, limitation calibration, failure attribution, change prediction, mission/identity lineage.
Circle / lambda	Substrate adapter with observe-propose-implement-validate-deploy-measure contract.	Circle-Bench: software/hardware/physical/biological/other substrate-specific closure tests.

13. Overall Scoring for One Harness-LLM Pair

13.1 Keep the Ordinal U-Level and Add a Continuous Pair Score

The highest certified Unified level U^* remains the primary ordinal statement. A continuous score is useful for comparing systems within the same level, measuring near-promotion progress, quantifying whether a harness change helped, and constructing a response curve across standardized harness generations. This paper therefore defines the Harness-LLM Intelligence Score, $HLIS(m,h)$, on a 0-100 scale for one frozen pair of base model m and harness h . The ordinal U -level and continuous $HLIS$ have different roles: U^* certifies qualitative maturity under cumulative gates, whereas $HLIS$ estimates balanced continuous strength inside the declared measurement profile.

$$HLIS(m,h) = 100 \times \left[\prod_{(d \in D)} A_d^{\alpha_d} \right]^{(1 / \sum_{(d \in D)} \alpha_d)}$$

An exactly equivalent log-space form, preferred for implementation because it is numerically stable, is:

$$HLIS(m,h) = 100 \times \exp\left\{ \left[\sum_{(d \in D)} \alpha_d \ln A_d \right] / \left[\sum_{(d \in D)} \alpha_d \right] \right\}.$$

13.2 Formal Definition and Meaning of the Symbols

Symbol	Meaning	Recommended interpretation
m	Frozen base LLM	The underlying model weights/version held fixed for the comparison.
h	Versioned harness	Prompting, memory, planning, tools, subagents, verification, learning, self-modeling, and other architecture around m .
$A(m,h)$	Realized agent/system	The concrete LLM+harness system actually evaluated.
D	Declared intelligence-family set	Singleton default: $\{C,I,DI,SA\}$. Organizational default: $\{C,I,O,DI,SA\}$.

Symbol	Meaning	Recommended interpretation
A _d	Normalized achievement in dimension d	Externally measured score in [0,1], constructed from versioned benchmark suites and cumulative level rules.
α_d	Predeclared dimension weight	Positive weight controlling the influence of dimension d. Equal weights are the default general comparison.
R	Resource envelope	Token, call, wall-time, compute, tool, memory, and external-access limits under which A _d is measured.
B	Frozen benchmark suite	Versioned public/hidden tasks, verifier code, and scoring definitions.

The more complete scientific notation is therefore HLIS(m,h;D, α ,R,B). The shorter HLIS(m,h) is used only when D, weights, resources, and benchmark versions are fixed by the evaluation protocol.

13.3 Choosing the Dimension Set D Without Double Counting

For a singleton agent, Organizational Intelligence is not assigned zero; it is not applicable and is excluded from D. A natural singleton set is D_{single} = {C,I,DI,SA}. For a true multi-agent organization, D_{org} = {C,I,O,DI,SA}. Delegation Intelligence is conceptually one family but is measured from T, H, and reliability p; therefore T and H should not also be inserted independently into the geometric mean when A_{DI} is already present. Doing so would double-count delegation.

$$D_{\text{single}} = \{C, I, DI, SA\} \quad \text{and} \quad D_{\text{org}} = \{C, I, O, DI, SA\}.$$

13.4 Constructing the Achievement Variables A_d

Each A_d must be an externally measured, normalized, versioned, and reproducible quantity in [0,1]. It should not be an LLM self-rating and should not be assigned informally by the harness being tested. A_d may be built hierarchically from several benchmark families, but the aggregation rule must be frozen before evaluation. Public benchmark results can contribute evidence, while higher I/O/SA claims require dedicated longitudinal or harness-state-grounded tests.

Achievement	Evidence source	Core construction rule
A _C	Reasoning/science/math/coding/abstract/multimodal/long-context benchmark families	Domain-balanced normalized aggregation rather than a single benchmark; long context does not certify persistent memory.
A _I	I0-IOmega longitudinal persistent-agent datasets, including long-term-memory subtests	Cumulative retention. Memory is embedded: I1 persistence/episodic continuity; I2 consolidation/procedural transfer; I3+ improvement/discovery lineage. No separate A _M is added.
A _O	O0-OOmega organizational datasets	Cumulative role/coordination/evolution tests with external verification.
A _{DI}	Delivered-correct surface S _A ^{del} (T,H), false-completion/false-rejection outcomes, and intervention logs	Publish F _A (H,p) from delivered-correct reliability. For continuous HLIS, optionally aggregate a predeclared loss-weighted net utility that penalizes false completions (and, if relevant, false rejections).

Achievement	Evidence source	Core construction rule
A_SA	SA0-SAOmega grounded self-model tests	Compare claims about self/state/limitations/change to harness or environment ground truth.

13.5 Why HLIS Uses a Weighted Geometric Mean

A simple arithmetic mean allows an exceptional score in one dimension to compensate too easily for a severe weakness in another. Unified Intelligence is intended to represent a balanced capability envelope, so the continuous pair score should preserve some bottleneck sensitivity even though hard promotion remains governed by U-level gates. The geometric mean supplies this property while remaining smooth and comparable within a declared dimension set.

Profile	Arithmetic mean	Geometric mean	Interpretation
[0.80,0.80,0.80,0.80,0.80]	0.80	0.80	Balanced capability.
[1.00,1.00,1.00,1.00,0.10]	0.82	≈ 0.63	One severe weakness is visibly penalized.

The geometric mean is not a replacement for the cumulative U gate. A system may have a comparatively high HLIS but remain at a lower U-level because one mandatory prerequisite is not satisfied. Conversely, a system can barely clear a higher U-level while having only moderate continuous strength. Reporting U* and HLIS together preserves both facts.

13.6 The Weight Parameters α_d

For a general leaderboard, equal α_d is the recommended starting point because it minimizes hidden value judgments and keeps systems comparable. Application-specific secondary scores may use different weights, but the weights must be declared before the hidden benchmark is run. They should never be adjusted after observing a model result.

$$\text{Equal-weight general score: } \alpha_C = \alpha_I = \alpha_O = \alpha_{DI} = \alpha_{SA} = 1.$$

For example, an AI4Science profile might legitimately emphasize C, I, and DI more than O if the evaluated subject is a single persistent research agent. Such a score should be labeled HLIS_AI4Science and reported beside, not instead of, the general score. This preserves a common headline comparison while allowing application-specific utility estimates.

13.7 Computing Cognitive Achievement A_C

A_C should represent broad cognitive capability rather than raw performance on one saturated benchmark. First normalize benchmark results within declared families, then aggregate within capability domains, and finally aggregate the domains. A possible domain vector is general reasoning, scientific reasoning, mathematics, abstract intelligence, coding, multimodal reasoning, and long-context cognition. Contemporary benchmarks such as HLE, GPQA, FrontierMath/AIME, ARC-AGI, LiveCodeBench, MMMU/MathVista, and RULER/LongBench can contribute evidence, but exact benchmark versions and normalization rules must be published. Long-context performance belongs here because it measures use of information made available within an episode; persistent cross-session memory is scored inside A_I instead.

$$A_C = G(R_{\text{general}}, R_{\text{science}}, R_{\text{math}}, R_{\text{abstract}}, R_{\text{code}}, R_{\text{multi}}, R_{\text{long}}).$$

A domain-balanced geometric or other predeclared aggregation is preferable to pooling all benchmark questions together, because otherwise a domain with many cheap items can dominate a domain represented by fewer but harder tasks.

13.8 Cumulative Achievement for I, O, and SA

Individual Intelligence, Organizational Intelligence, and Self-Awareness are explicitly hierarchical. Let $q_{(d,k)}$ be the externally verified performance on the level- k test suite for dimension d . To prevent level skipping, define cumulative performance at level k by the weakest required lower stage:

$$q^*_{(d,k)} = \min_{\{j \leq k\}} q_{(d,j)}.$$

A continuous achievement score can then aggregate $q^*_{(d,k)}$ across levels with predeclared level weights w_k :

$$A_d = [\sum_k w_k q^*_{(d,k)}] / [\sum_k w_k], \quad d \in \{I, O, SA\}.$$

This construction means that a system cannot receive full continuous credit for I4-like behavior if it fails an I2 retention requirement. Higher-level evidence can still provide partial information, but the cumulative envelope remains limited by lower-stage reliability. The same principle implements the paper's central rule $I_{(k+1)} \Rightarrow I_k$, $O_{(k+1)} \Rightarrow O_k$, and $SA_{(k+1)} \Rightarrow SA_k$.

For A_I specifically, the level suites must embed progressively deeper long-term-memory requirements rather than award a separate A_M . I1 tests persistence, episodic/task-state recovery, provenance, temporal ordering and stale-state handling; I2 adds semantic consolidation, knowledge updating, proceduralization and held-out transfer; I3 adds durable failure/hypothesis/change/benchmark/promotion lineage; I4 adds meta-improvement history; I5 adds hypothesis-experiment-evidence research memory; IOmega may test governed evolution of memory mechanisms themselves. Because these memory functions are prerequisites inside I, adding A_M to the top-level HLIS product would double-count the same capability.

A memory benchmark should score usefulness, not storage volume. Recommended diagnostics include relevant-recall precision/recall, multi-session reasoning, temporal reasoning, knowledge update and contradiction handling, abstention when evidence is absent, provenance accuracy, consolidation quality, procedural transfer, repeat-error reduction, stale-memory intrusion, and downstream task gain attributable to retrieved memory. LongMemEval, LoCoMo, and related long-term-agent-memory benchmarks can provide external evidence for parts of I1-I2, while HIL-native longitudinal suites are still required for higher I-levels [21-24].

13.9 Delegation Achievement A_{DI} : Capability, Delivery Integrity, and Human Load

Delegation Intelligence has three related but non-identical observables: task capability, delivery integrity, and human load. The certification frontier is based on delivered-and-correct completion $S_A^{\text{del}}(T,H)$. Outcome integrity is measured by false completion and false rejection. Human cognitive load remains represented by H , Critical Intervention Depth, intervention count/minutes, and any escalation required after held-back work.

$$F_A(H,p) = \max \{ T : S_A(T,H) \geq p \}.$$

13.9.1 Why a success-only aggregate is insufficient

If two paired harnesses differ only by a post-hoc acceptance step, the external verifier verdict on the produced artifact is unchanged. Therefore a gross score defined only as $P(V=1)$ is invariant to acceptance even when false completions fall sharply. This is not a defect in verifier accuracy; it is a mismatch between the scoring variable and the delegator-facing outcome.

Proposition 1 (paired acceptance invariance). For the same executor artifact, adding a purely post-hoc accept/hold-back step cannot change $P(V=1)$. Any gross delegation aggregate built only from verifier-pass probability is therefore identical across those paired rungs. A benchmark that wants to value outcome integrity must record delivery/acceptance separately.

13.9.2 Outcome-complete delegation scoring

For every attempt record D (delivered/accepted) and V (external verifier pass). This yields four outcomes: delivered-correct ($D=1, V=1$); false completion ($D=1, V=0$); held-back invalid work ($D=0, V=0$); and false rejection ($D=0, V=1$). The first is successful delegation; the second is the most important reliability failure; the third is a non-completion that should appear in human-load/escalation metrics; and the fourth measures excessive strictness.

A loss-weighted net surface may be used as a secondary continuous scoring primitive:

$$S_A^{net}(T,H; \lambda_{FC}, \lambda_{FR}) = P(D=1, V=1) - \lambda_{FC} P(D=1, V=0) - \lambda_{FR} P(D=0, V=1).$$

The loss multipliers λ_{FC} and λ_{FR} are application/task-class parameters and must be fixed before evaluation. If false rejection has negligible cost, λ_{FR} may be zero. Unlike the earlier success-only aggregate, the positive term is delivered-and-correct rather than verifier-pass, so a correct result that the harness refuses to deliver is not credited as successful delegation.

$$A_DI = [\sum_{(t,h)} w_t v_h \text{clamp}(S_A^{net}(T_t, H_h), 0, 1)] / [\sum_{(t,h)} w_t v_h].$$

Task weights w_t increase with difficulty and intervention weights v_h increase as required human cognition decreases. One illustrative intervention schedule is $H_0=6, H_1=5, H_2=4, H_3=3, H_4=2, H_5=1$; all such weights and loss ratios are calibration choices, not laws. The scalar A_DI must never replace the frontier: at minimum publish $F_A(H_0, p)$, $F_A(H_1, p)$, $F_A(H_2, p)$, false-completion rate, false-rejection rate, and human-load statistics.

13.10 Numerical Example of HLIS

Consider an organizational harness-LLM pair with equal weights and normalized achievements $A_C=0.90$, $A_I=0.70$, $A_O=0.60$, $A_DI=0.80$, and $A_SA=0.75$. Then:

$$HLIS = 100 \times (0.90 \times 0.70 \times 0.60 \times 0.80 \times 0.75)^{(1/5)} \approx 74.3.$$

Dimension	Achievement
C	0.90
I	0.70
O	0.60
DI	0.80
SA	0.75

The recommended report is not merely “HLIS=74.3.” It should preserve the discrete profile and certification context, for example: U3 | HLIS 74.3 | [C5,I3,O3,T4/H1,SA3] | p=0.80 | false-completion

rate=0.00 | benchmark version Bv1.5 | ladder id=L-HIL-1 | resource envelope R-Standard. This allows a reader to see both the continuous aggregate and the source of the score.

13.11 Zero Scores, N/A Dimensions, and Hard Gates

A zero in a required geometric-mean component forces HLIS to zero. This can be appropriate if the system completely fails a foundational required dimension, but it must not be confused with “not applicable.” For a singleton system, O is omitted from D rather than set to zero. If a benchmark produces a true zero but a continuous nonzero pair score is analytically desired, a small preregistered floor ϵ may be used only as a secondary sensitivity analysis; the primary U-level gate should remain strict. This paper therefore recommends hard ordinal gates plus a soft continuous HLIS, not replacing the gate with numerical smoothing.

13.12 Run Logs Must Record Acceptance and Delivery Outcomes

A run log that stores only `verifier_pass` cannot distinguish a wrong artifact that was confidently delivered from one that an acceptance mechanism correctly held back. Formal HIL runs should therefore record `verifier_pass` and `harness_accepted/delivered` separately. For harnesses without an acceptance step, `harness_accepted` is recorded as null and delivery is treated as true by convention.

Required field	Meaning
<code>verifier_pass</code>	External ground-truth/check verdict on the artifact.
<code>harness_accepted / delivered</code>	Whether the tested harness returned the artifact as completed work.
<code>false_completion</code>	Derived: <code>delivered=1</code> and <code>verifier_pass=0</code> .
<code>held_back_invalid</code>	Derived: <code>delivered=0</code> and <code>verifier_pass=0</code> .
<code>false_rejection</code>	Derived: <code>delivered=0</code> and <code>verifier_pass=1</code> .
<code>attempts</code>	Number of executor attempts consumed.
<code>verification_responsive</code>	Whether a verifier rejection was converted into a later verified correction without human help deeper than CID1.

13.13 Reliability, Confidence Intervals, and Resource Envelopes

Every achievement estimate is sampled from tasks and therefore uncertain. HLIS should eventually be accompanied by a confidence interval, for example $HLIS=74.3 \pm 1.8$ or a bootstrap 95% interval. The same benchmark tasks should be rerun across seeds or matched task families when stochasticity is material. A one-off success at a high level is not equivalent to reliable capability.

Resource conditions must also accompany the score. The complete notation $HLIS(m,h;D,\alpha,R,B)$ makes explicit that a 500-call multi-agent harness and a one-call harness are not automatically comparable if their resource envelopes differ. Recommended R fields include total model tokens, number of LLM calls, wall-clock time, compute class, persistent-memory budget, tool/API access, external knowledge access, and number of subagents. Cost and latency should be reported beside intelligence rather than silently included in it. A separate efficiency or Pareto analysis can compare intelligence against cost and time.

13.14 HLIS Is the Building Block of HIL

HLIS evaluates one point: one frozen LLM under one harness. HIL evaluates the same frozen LLM across a standardized ladder of harness generations. If HG_k is the k th standardized harness, the Harness Scaling Curve is:

$$HSC_m(k) = HLIS(m, HG_k).$$

The sequence $\{HLIS(m, HG_0), \dots, HLIS(m, HG_K)\}$ is then used to compute HIL-Level, HIL-AUC, HIL-Ceiling, and Harness Gain. In this sense, HLIS is to one harness-LLM system what an individual point is to a scaling curve; HIL characterizes the model's ability to convert increasingly capable harness structure into verified intelligence.

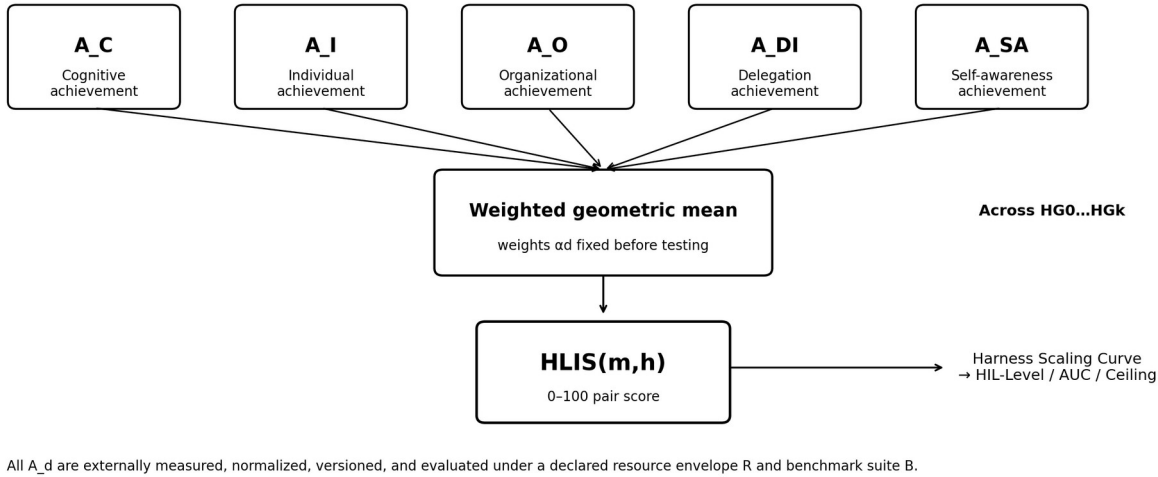


Figure 9. HLIS aggregates externally measured intelligence-family achievements for one LLM-harness pair. Repeating the pair evaluation across a standardized harness ladder produces the Harness Scaling Curve and the higher-level HIL characterization.

13.15 Recommended Pair-Level Reporting Standard

Field	Example
System	Model-X + HG3-v1.0
Unified level	U3
HLIS general	74.3 [95% CI 72.7-75.9]
Core profile	C5 I3 O3 DI frontier H0:T3, H1:T4 SA3
Reliability	$p^*=0.80$ certification frontier
Dimension set / weights	$D=\{C,I,O,DI,SA\}$; equal α_d
Benchmark suite	Bv1.3; hidden verifier set frozen before test
Resource envelope	R-Standard: declared tokens/calls/time/tools/subagents
Efficiency context	Cost/task and latency/task reported separately

This reporting standard prevents HLIS from becoming an opaque single number. A pair report is incomplete unless it carries U^* , HLIS with dimension set and weights, the discrete profile, the named bottleneck, reliability with an interval, the delegation frontier, false-completion and false-rejection rates, human intervention/CID, the resource envelope, benchmark version, and ladder identifier.

14. Harness-Intelligence-Level (HIL) Characterization of a Base LLM

14.1 Why an LLM Needs a Harness-Based Score

A base LLM should not be judged only in a minimal chat harness, because some models may be unusually capable of using memory, planning, tools, multiple roles, verification, and self-improvement machinery. Conversely, an elaborate harness can mask a model that contributes little. To measure the

ability of one LLM to express intelligence through harnesses, evaluate the same frozen model across a standardized harness ladder $H = \{HG0, HG1, \dots, HG6, HG\Omega\}$. Tools, external knowledge access, resource ceilings, authority, and hidden benchmarks should be standardized so the comparison reflects model-harness interaction rather than uncontrolled infrastructure advantages.

14.2 HIL-Level, HIL-AUC, HIL-Ceiling, and Harness Gain

Metric	Definition	Interpretation
HIL-Level(m)	Highest standardized harness generation k for which $A(m, HG_k)$ satisfies the corresponding cumulative target gate.	How far up the harness ladder the LLM can validly exploit the standardized architecture.
HIL-AUC(m)	Weighted mean of $HLIS(m, HG_k)$ across all tested standardized harness generations.	Broad harness-expressible ability; rewards models that work well across multiple harness complexities.
HIL-Ceiling(m)	$\max_k HLIS(m, HG_k)$.	Best intelligence score that a standardized harness can realize from the model.
Harness Gain(m)	$\max_k HLIS(m, HG_k) - HLIS(m, HG_0)$.	How much capability is unlocked by harnessing beyond the minimal baseline.
Harness Design Score (HDS)	Mean independently verified gain produced when the LLM proposes candidate harness improvements under a fixed design budget.	Ability of the LLM to improve the architecture that harnesses it; reported separately from baseline HIL.
Verification Responsiveness VR	Fraction of verifier-rejected attempts converted into a later verified accepted correction without human help deeper than CID1.	How effectively the model-harness pair uses external failure evidence; feedback protocol must be standardized.

14.3 Verification Responsiveness VR

Harness Gain says how much a richer harness helped, but not which part required the model. Acceptance and rollback can reduce harmful delivery even if the model contributes nothing after its first attempt. Retry, replanning, and routing require the executor to convert detected failure into a correction. Verification Responsiveness measures that interaction directly.

$VR(m, h) = (\# \text{ verifier-rejected attempts followed by a verified accepted correction with no human help deeper than CID1}) / (\# \text{ verifier-rejected attempts eligible for retry})$.

VR must be reported with the granularity and correctness of the rejection feedback. A model cannot respond to a misleading or underspecified failure message; therefore VR is a pair-level interaction diagnostic with a model contribution, not a pure intrinsic-model scalar. It should be measured under standardized feedback protocols when comparing LLMs.

14.4 A Single HIL Score and Structural Cautions

A single HIL-Score may be published as a secondary convenience, but Version 1.5 treats the Harness Scaling Curve and a compact summary tuple as primary. HIL-AUC and HIL-Ceiling are structurally correlated because Ceiling is the maximum of the same values that AUC averages; moreover, AUC can rank a high-baseline flat model above a lower-baseline steep model even when harness response is the property of interest. Any composite weights therefore require empirical calibration against downstream outcomes rather than aesthetic choice.

Recommended primary summary: <HIL-Level, HLIS(HG0), HIL-Ceiling, Harness Gain, VR, ladder id>, with the complete Harness Scaling Curve published beside it. If the earlier illustrative composite HIL-Score = $0.55 \cdot \text{AUC} + 0.35 \cdot \text{Ceiling} + 0.10 \cdot \text{Harnessability}$ is retained for continuity, label it provisional and publish every component.

A high HIL-Score therefore means more than “the model answers hard prompts.” It means that, across a fixed suite of increasingly capable harnesses, the LLM repeatedly converts architectural support into externally verified intelligence. This makes HIL a candidate metric for comparing the harness-expressible ability of different LLMs.

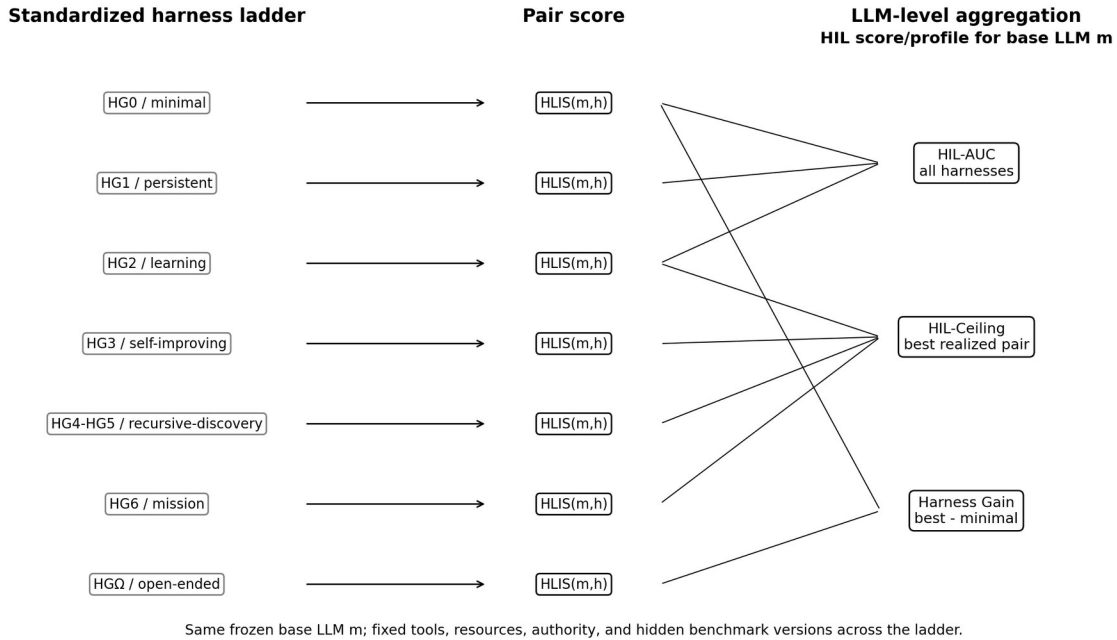


Figure 6. The same base LLM is tested across a standardized harness ladder. Each pair receives HLIS; the collection of pair scores yields HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, and an optional composite HIL-Score.

14.5 Recommended LLM Scorecard

Score	Subject being measured	Primary use
C-Score / model baseline	Base LLM in a minimal standardized harness.	Raw/broad cognitive problem-solving comparison.
Delegation Surface / DI score	One LLM-harness pair.	How hard a task can be delegated at each human-intervention budget.
Unified level U*	One LLM-harness pair or organization.	Highest cumulative intelligence stage actually certified.
HLIS	One frozen LLM-harness pair.	Continuous 0-100 comparison within/across levels.
HIL-Level + Harness Scaling Curve	One base LLM across the standardized harness ladder.	How much intelligence the LLM can express across standardized harnesses; report baseline, ceiling, gain, VR, and optional composite.
HDS / Harness Design Gain	LLM acting as a harness designer under frozen external evaluation.	How well the model can design architectures that improve its own or another model's realized intelligence.
Verification Responsiveness VR	One base LLM measured through standardized	Whether detected failure is converted into verified

Score	Subject being measured	Primary use
	retry/feedback harnesses.	correction; separates passive acceptance gains from model-responsive harness gains.

15. Benchmark Library for Harness-Level Intelligence

A harness intelligence program should maintain a versioned benchmark registry rather than one monolithic test set. Every level has a harness contract and a matched dataset. The minimal unit is a task manifest containing target coordinate/level, frozen model and harness hashes, tool/resource/authority budget, intervention policy, hidden verifier, success threshold, and required retention suites. The I-Bench registry should contain a dedicated longitudinal-memory sub-suite rather than a separate top-level intelligence family.

15.1 Task Binding Status

A registry of task specifications is not automatically an executable benchmark. Every task row should record one of three statuses: bound (a runnable instance and verifier exist for this family/band); band-only (some runnable task exists at the same difficulty band but not this family); or specification-only (the task is defined but not instantiated). Band-only must not be counted as coverage, because substituting another domain at the same T band changes the construct being measured.

The longitudinal-memory sub-suite should include: cross-session persistence and restart recovery; relevant memory selection under distractor load; temporal ordering and current-vs-obsolete state; knowledge updates and contradiction resolution; abstention when memory evidence is missing; provenance tracing; semantic consolidation; proceduralization and transfer; memory-induced repeat-error reduction; and, at higher I-levels, self-improvement/version, meta-improvement, and discovery lineage. Formal runs should include genuine process/context discontinuities so that copying the entire transcript into a long context cannot masquerade as persistent memory.

Suite	Purpose	Example level-specific evidence
DLI-Bench	First-stage delegation surface.	T0 atomic execution; T2 multi-step persistence; T3 strategy construction; T4 long-horizon project; T5 hidden solution; T6 mission; Tomega charter cycles, each under H budgets.
U-Bench-U0...UOmega	Unified promotion and lower-level retention.	At Un, rerun U0...U(n-1) retention plus the new gate for C/I/O/DI/SA.
I-Bench-I0...IOmega	Persistent individual evolution with long-term memory embedded as the temporal substrate.	I1 multi-session recall/state recovery/provenance; I2 consolidation, update handling and procedural transfer; I3 change/version lineage; I4 meta-improvement history; I5 discovery memory; IOmega memory-mechanism evolution lineage.
O-Bench-O0...OOmega	Collective intelligence.	Routing, persistent organization, adaptive coordination, reorganization, meta-improvement, collective discovery, new-role creation.

Suite	Purpose	Example level-specific evidence
SA-Bench-SA0...SAOmega	Evidence-grounded self-awareness.	State truth, capability limits, causal self-diagnosis, predicted self-change, self-unknowns, mission identity, lineage.
C-Bench	Broad cognitive capability.	Multiple reasoning/domain families to avoid mistaking one specialty for general cognition; long-context handling is a C subdomain, not an I1 memory certification.
Circle-Bench	Application/substrate reach, reported orthogonally.	Software closure first; later hardware/physical/biological/other substrate-specific implementation and validation loops.

This organization supports the proposed development workflow: start with Delegation Intelligence because it is immediately operational; then certify a Unified level; then add the coordinate suites most important for the target application. A research worker might emphasize I and SA, a fleet O and DI, an LLM benchmark C and HIL, and a physical autonomy program DI plus Circle completion. The common scores remain comparable because the application panels are reported in addition to, not instead of, the standardized core.

16. Relationship to Contemporary Model and Agent Benchmarks

16.1 Existing Benchmarks Measure Important Slices, Not a Single Intelligence Quantity

Modern model evaluation is intentionally plural. HLE and MMLU-Pro emphasize broad difficult academic reasoning; GPQA targets graduate-level scientific questions; FrontierMath and competition mathematics target advanced mathematical reasoning; LiveCodeBench and SWE-bench probe coding and repository-level software work; ARC-AGI emphasizes fluid adaptation on novel tasks; BFCL evaluates tool/function calling; the tau-bench family evaluates interactive tool-agent-user reliability; BrowseComp and GAIA probe browsing and general assistant behavior; MMMU and MathVista evaluate multimodal reasoning; RULER and LongBench probe long-context behavior; SimpleQA targets factuality; Chatbot Arena measures human preference; and OSWorld evaluates multimodal computer-use agents [3-20]. These benchmarks are complementary because they deliberately stress different capabilities. LongMemEval and LoCoMo explicitly probe long-term conversational memory, while LongMemEval-V2 extends the question toward agent experience in specialized environments [21-24]. These are especially relevant to the I1-I2 evidence layer, but they still do not by themselves establish governed self-improvement or recursive evolution.

HIL therefore should not be presented as replacing these scores. Let $B(m)$ denote the vector of contemporary benchmark results for base model m . The HIL program consumes $B(m)$ as part of the evidence layer, then asks a different question: how does the same frozen model behave when embedded in progressively stronger, standardized harness architectures? The central distinction is benchmark slice versus harness-expressible system intelligence.

16.2 Mapping Contemporary Benchmarks into the HIL Measurement Architecture

Capability slice	Representative benchmark families	Primary HIL evidence	What the benchmark alone does not establish
Broad reasoning	HLE; MMLU-Pro	C baseline / breadth and hard closed-ended reasoning	Persistence, autonomy, organization, self-improvement, or harness scaling.
Scientific reasoning	GPQA Diamond	C-science; expert scientific reasoning evidence	Longitudinal scientific learning or autonomous research execution.
Advanced mathematics	AIME-style competition math; FrontierMath	C-math; difficult and frontier mathematical reasoning	Project autonomy, tool orchestration, or learning across episodes.
Coding	LiveCodeBench	C-code plus bounded execution evidence	Repository-scale planning, persistence, human intervention, or organization.
Software engineering	SWE-bench and newer repo-level suites	DI at T2-T4; C-code; execution/verifier evidence	A general intelligence level unless human intervention and cross-domain retention are controlled.
Abstract / adaptive intelligence	ARC-AGI-2/3 family	C-fluid; DI exploration/adaptation; efficiency evidence	Persistent individual evolution, organizational evolution, or broad real-world coverage.
Tool calling	BFCL	DI tool-selection/action layer; abstention and multi-step evidence	Long-horizon mission autonomy by itself.
Interactive agent reliability	tau-bench family and maintained descendants	DI success surface; reliability p; policy/tool interaction	General cognition or self-improvement depth.
Research / browser agent	BrowseComp; GAIA	DI for research tasks; C retrieval/reasoning; tool reach	Longitudinal learning, organizational evolution, or self-model depth.
Vision + reasoning	MMMU; MathVista	C-multimodal / visual reasoning	Autonomous action, persistence, or harness scaling.
Long context	RULER; LongBench	C-context: retrieval and reasoning over information present within the evaluation episode	Persistent cross-session memory, restart recovery, consolidation, provenance, or useful learning over time; those belong to I.
Factuality	SimpleQA and related factuality suites	C-factuality / calibration evidence	Autonomy, organization, or self-improvement.
Real-user preference	Chatbot Arena / human preference	External preference / usability evidence; useful secondary panel	A mechanistic or cumulative intelligence level.
Computer agent	OSWorld family	DI in executable GUI environments; multimodal C; task completion	Longitudinal I/O/SA maturity unless separately instrumented.
Long-term memory / experience	LongMemEval; LoCoMo; LongMemEval-V2; memory-harness evaluations such as MemGPT-style multi-session settings	I1 persistence/episodic retrieval; portions of I2 temporal reasoning, knowledge update, and experience use	Higher I2 consolidation/procedural learning, I3+ self-improvement lineage, or Unified level unless separately tested.

The mappings are deliberately many-to-many. A benchmark can contribute evidence to more than one coordinate, but no benchmark is allowed to certify an unrelated coordinate merely because it is difficult. Reports must name the exact benchmark version, task split, harness, tool/resource budget, and evaluation date because benchmark families evolve over time. For example, BFCL and the tau-bench line have evolved across versions and increasingly agentic settings; HIL should therefore cite frozen benchmark versions rather than benchmark names alone [10-11].

Contemporary benchmarks are evidence slices; HIL adds harness response and longitudinal structure.

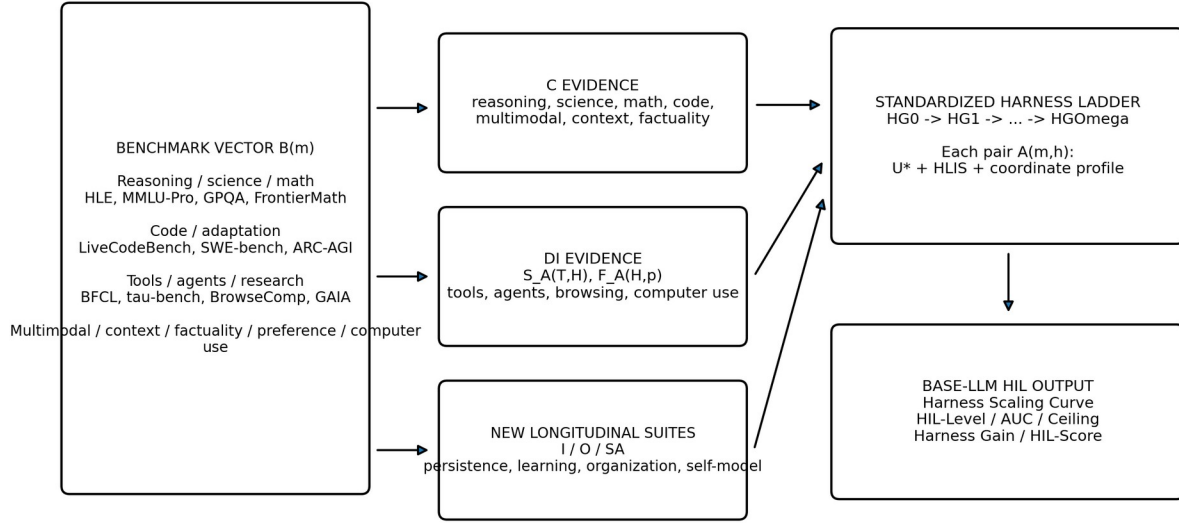


Figure 7. Contemporary benchmarks remain the evidence layer. HIL organizes their evidence into cognitive, delegation, and longitudinal/structural coordinates, then measures how those capabilities compose under a standardized harness ladder.

16.3 The Main Added Quantity: Harness Response

Most benchmark leaderboards primarily compare systems at one evaluation configuration. HIL adds an explicit response-to-harness dimension. For frozen model m and standardized harness generation HG_k , define $HSC_m(k) = HLIS(m, HG_k)$. The ordered set $HSC_m = \{(k, HLIS(m, HG_k))\}$ is the Harness Scaling Curve. It reveals whether model capability continues to convert into higher realized intelligence as persistence, tools, delegation, learning, organization, verification, and self-improvement machinery are added.

Two models can therefore reverse order as harness complexity increases. A model with a stronger minimal-harness benchmark profile may plateau early, while another with a weaker HG_0 score may exploit persistent long-term memory, planning, verifier feedback, multi-agent protocols, and long-horizon state much more effectively. HIL-AUC, HIL-Ceiling, Harness Gain, and HIL-Level summarize this curve, but the curve itself should remain a primary result because it exposes saturation and non-monotonic behavior.

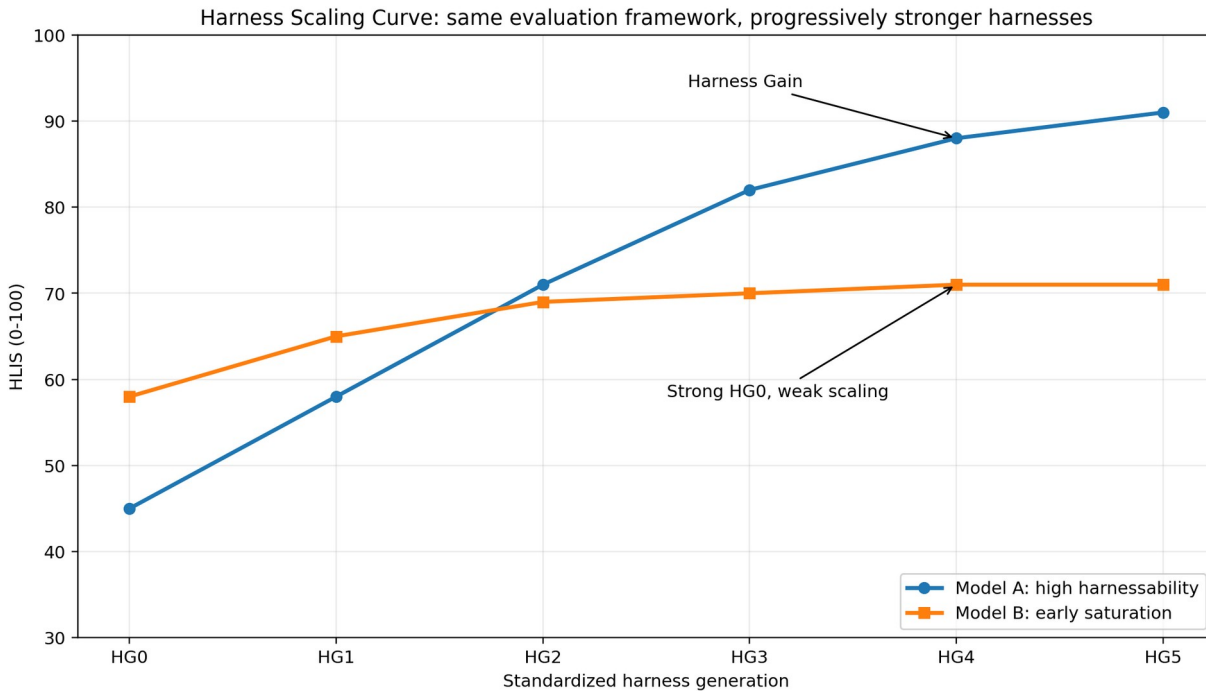


Figure 8. Illustrative Harness Scaling Curves. Model B is stronger in the minimal harness but saturates; Model A converts stronger harness generations into substantially higher realized intelligence. The curves are hypothetical and illustrate the measurement concept, not empirical model results.

16.4 Pilot Reference Measurement: What Building the Ladder Revealed

A preliminary reference implementation built HG0-HG3 for Delegation Intelligence and ran three executors from two vendor families under a fixed H1 intervention budget. The study is intentionally small and is not presented as a product ranking. Its value is methodological: it exposed a scoring invariance and several experimental-design requirements that were not obvious from the equations alone.

16.4.1 Paired acceptance invariance

For paired HG0/HG1 outputs, gross verifier success is exactly identical by construction because the executor artifact is the same. A separate acceptance process can nevertheless convert false completions into held-back invalid work. In the controlled paired check, gross success stayed fixed while the net outcome score improved and false rejections were zero. This is the empirical motivation for separating verifier verdict from delivery/acceptance in Section 13.9.

Executor	Gross HG0	Gross HG1	Net HG0	Net HG1	False rejections
Family A, stronger	0.933	0.933	0.867	0.933	0
Family A, weaker	0.917	0.917	0.833	0.917	0

16.4.2 Harness response, saturation, and cost

Across the initial four-rung pilot, stronger frontier executors had little headroom and produced nearly identical aggregate curves, while a weaker executor gained more from retries and harness support. Identical aggregate curves did not imply equivalent executors: they failed different instances and differed materially in wall-clock cost. The lesson is that a Harness Scaling Curve is only as

discriminative as its task-suite headroom, and intelligence scores must be reported beside resource envelopes and per-instance failure patterns.

16.4.3 Paired-rung protocol

Whenever two rungs do not alter the executor request and differ only in post-hoc acceptance or evaluation machinery, run the executor once per task/seed and score both rungs from the same artifacts. Independent reruns introduce stochastic variance that can be larger than the mechanism under study. The paired design is both cleaner and cheaper.

16.5 Why HIL Adds Information Beyond a Benchmark Average

Property	Typical benchmark score	HIL extension
Unit of analysis	Model or agent in one benchmark configuration.	Frozen LLM-harness pair A(m,h), plus the same LLM across a harness ladder.
Capability breadth	A vector of specialized scores.	Preserves the vector but organizes it into C/DI/application evidence and adds I/O/SA longitudinal tests.
Autonomy	Sometimes implicit or benchmark-specific.	Explicit T x H x p delegation surface and frontier F_A(H,p).
Human dependence	Often not a first-class reported variable.	H0-H5 cognitive intervention is logged and constrained.
Persistence / learning	Usually one episode or fixed model state.	I-level tests embed long-term memory: real restart continuity, episodic retrieval/provenance, temporal update handling, consolidation/procedural transfer, and later governed self-improvement lineage.
Organization	Often a fixed agent architecture.	O-level tests evaluate coordination, reorganization, and collective improvement with real role boundaries.
Self-awareness	Usually absent or conversational.	SA requires evidence-grounded state, limitation, causal-failure, and change-lineage tests.
Model versus harness progress	Usually conflated in the final system score.	HLIS measures one pair; Harness Gain and HIL separate model baseline from harness unlock.
Higher-order improvement	Not normally measured.	HG3+ can test whether the system improves cognition/organization under frozen external evaluation.

The strongest claim for HIL is therefore not that it produces one universally superior scalar. Its value is that it measures a new response function: how much externally verified intelligence a base LLM can express as the harness around it becomes more capable. The overall HIL-Score is a convenience summary; the benchmark vector B(m), Delegation Surface, Unified level, coordinate profile, and Harness Scaling Curve remain the primary scientific outputs.

16.6 Recommended Two-Layer Scorecard for Frontier LLMs

Layer	Report	Purpose
Contemporary benchmark layer	HLE/MMLU-Pro; GPQA; math; coding/SWE; ARC-AGI; BFCL; agent/research; multimodal; long context; factuality; human preference; computer-use benchmarks, with exact versions.	Shows the familiar capability slices and allows comparison with existing leaderboards.

Layer	Report	Purpose
HIL layer	C profile; $S_A^{\Delta}(T,H)$, $F_A(H,p)$, false-completion/rejection rates; I/O/SA; U*; per-harness HLIS; HSC; HIL-Level; baseline; Ceiling; Gain; VR; optional calibrated composite.	Shows how the model becomes a persistent, delegated, organizational, self-modeling system as harness sophistication increases.

This two-layer reporting strategy avoids the central failure mode of a single leaderboard number. A model can remain visibly strong in science but weaker in coding or tool use, while HIL separately reveals whether those abilities compose into reliable autonomous systems. For real deployments, the application-specific coordinate panel from Section 12.3 is then added without altering either layer.

16.7 Relationship to Prior AGI-Level Proposals

The proposed hierarchy is related to, but distinct from, prior AGI taxonomies. Morris et al. separate performance, generality, and autonomy and argue for staged operational definitions rather than a binary AGI label [1]. Hendrycks et al. define AGI in terms of the versatility and proficiency of a well-educated adult across a psychometrically grounded set of cognitive domains [2]. The present framework preserves broad cognitive evidence in C, but adds persistent individual evolution I, organizational evolution O, operational self-awareness SA, the explicit Delegation Surface (T,H,p), and a standardized harness-response characterization. HIL is therefore best viewed as a harness-centric complement to capability- and human-reference-centered AGI definitions rather than a replacement for them.

17. Benchmark and Promotion Protocol

- Freeze the tested model, harness, prompts, tool set, authority profile, resource envelope, benchmark version, and ladder identifier before certification.
- When two rungs differ only in post-hoc machinery, use the same executor artifacts for both rungs; do not introduce unnecessary between-rung stochasticity.
- Record verifier_pass and harness_accepted/delivered separately so false completions, held-back invalid work, and false rejections remain observable.
- Publish task binding status (bound, band-only, specification-only) and do not count band-only tasks as coverage for an uninstantiated family.
- For harness-design experiments, the designer model may not alter or select the hidden certification set, verifier, or promotion criterion.
- Freeze the tested harness, model, tool set, prompts, and authority profile before certification.
- Use multiple task families per T band so difficulty is not synonymous with one domain such as software.
- Run at predeclared intervention budgets H0-H5 with a written human-assistance policy.
- Use an independent verifier or sealed reference; the performing agent cannot certify its own success.
- Record cognitive interventions separately from governance approvals.
- Record Critical Intervention Depth so a single human-supplied central strategy cannot be hidden by a low intervention count.
- Use confidence intervals and repeated tasks to estimate $S_A(T,H)$, then report the frontier $F_A(H,p)$.
- For U promotion, rerun a retention suite for all lower levels and then run the new-level suite.
- For I3+, O3+, SA4+, and U4+, require longitudinal evidence across multiple self/organizational changes rather than a single successful modification.

- For T5+, use sealed novelty environments, hidden mechanisms, or procedurally generated certification tasks to reduce memorization and benchmark leakage.
- For I1+, include true episode boundaries or process/context restarts; do not allow the full prior transcript to remain in the active context when the test is intended to measure persistent memory.
- For I1-I2 memory tests, measure relevance, provenance, temporal updates, contradiction handling, abstention, and downstream usefulness in addition to raw recall; a larger memory store is not automatically a better memory system.
- For I2+, require held-out behavioral transfer from consolidated experience; for I3+, preserve externally auditable change/version lineage so later self-improvement claims can be reconstructed from durable evidence.

18. What Should Remain Outside the Unified Core

Not every important property should be renamed as another intelligence. Several variables are essential for interpreting a system but are better treated as structural, environmental, or substrate coordinates.

Coordinate	Why it remains outside core intelligence
Circle / λ	Measures which substrate improvement loops are actually reached or closed; substrate access is not identical to cognition.
Write depth D	Measures which persistent cognitive states can be rewritten; permission to rewrite does not prove intelligent rewriting.
Authority A	Permission is not intelligence. A highly intelligent verifier may deliberately have almost no action authority.
Reach R	Access to files, tools, agents, robots, labs, or networks increases possible action but is not itself cognition.
Verification V	Determines credibility/independence of claims; it validates intelligence rather than constituting the tested intelligence.
Separation S	Organizational validity condition for proposer/evaluator/promoter roles.
Resources	Compute, memory, money, time, energy and hardware affect achievable performance but should not inflate the intelligence definition.
Physical floors ϕ	Represents irreducible latency/physics constraints and degree of saturation rather than cognitive level.

Layered description of an intelligent system

Unified headline	U
Core intelligence	C, I, O, DI, SA
Measured delegation	T, H, p / frontier F_A(H,p)
Substrate reach	$\lambda = (\lambda I, \lambda II, \lambda III, \lambda IV, \lambda V+)$
Structural validity	D, A, R, V, S
Resources / floors	compute, memory, time, energy, ϕ

Figure 4. A complete system description layers the Unified headline over the core intelligence profile, measurable delegation frontier, substrate reach, structural validity conditions, and resource/physical constraints.

19. Worked Examples

19.1 Strong LLM, weak harness

[C5, I0, O0, T2, H2, SA1]

The model may solve difficult reasoning problems, but the instantiated agent has little persistent evolution and limited end-to-end delegation autonomy. High C does not automatically produce high U.

19.2 Moderate model, strong persistent harness

[C3, I2, O1, T4, H1, SA2]

This system can be highly useful operationally: persistence, planning, tools, retries, and verification allow difficult tasks to be delegated. Yet it is not self-improving in the I3 sense, and its self-model is not yet causal. The profile communicates these strengths and limits directly.

19.3 Advanced organization with a bottleneck

[C5, I4, O3, T4, H1, SA4] -> U3, bottleneck O3

Several dimensions exceed level three, but organizational intelligence has not reached O4. The unified level remains U3 until the organization demonstrates recursive improvement while retaining lower-level capabilities.

19.4 Same LLM Across a Harness Ladder

A base LLM is tested in HG0, HG1, HG2, and HG3. Its HLIS scores are 42, 55, 67, and 72; the corresponding certified levels are U0, U1, U2, and U3. The model therefore has HIL-Level 3 on the current standardized ladder, a ceiling of 72, and a large positive Harness Gain of 30 points. A second model might start at HLIS 60 in HG0 but rise only to 70 in HG3: it has stronger minimal-harness capability but lower harness gain. HIL-AUC and HIL-Ceiling distinguish these patterns rather than collapsing them into one prompt benchmark.

20. Implications for Intelligence Research

A practical consequence is that frontier-model evaluation should retain contemporary benchmark vectors rather than compress them prematurely. HIL adds a second axis of progress: harness scaling. This allows researchers to distinguish a new model that improves raw reasoning from a new harness that unlocks more of an existing model, and to identify models whose strengths are unusually composable into persistent agents or organizations.

The framework distinguishes model capability from agent capability and both from system capability. A benchmark that tests only answers under carefully prepared prompts primarily measures C. A benchmark that tests whether the same system can accept a difficult goal, maintain state, choose strategy, use tools, recover, verify, and finish with little human cognition measures the delegation surface. Longitudinal tests of learning and self-modification measure I, with long-term memory embedded as the temporal substrate: cross-session continuity at I1, consolidation and procedural transfer at I2, and improvement/discovery lineage at higher levels. Multi-agent role and reorganization tests measure O. Grounded introspective tests measure SA.

The pilot ladder adds a measurement principle: a score defined only on “how often was the produced artifact correct?” can miss differences that matter to delegation, such as whether incorrect work was confidently returned as finished or correctly held back. Outcome-complete evaluation should therefore preserve both task capability and delivery integrity rather than forcing them into one undifferentiated success bit.

This decomposition also reduces incentives to overclaim. A system can legitimately report high performance in one dimension without claiming a uniformly high intelligence level. Conversely, the cumulative U hierarchy creates a demanding standard for statements such as 'U4 intelligence': the system must retain lower capabilities and satisfy all required gates at the specified reliability.

The framework is intentionally extensible. Empirical work may revise thresholds or reveal that a currently grouped construct should split. However, new axes should be added only when a property is both operationally measurable and not adequately explained by the existing core or by structural/environmental coordinates.

The HIL framework also creates a new object of study: harnessability. Two LLMs with similar C-scores may differ substantially in their ability to exploit persistent memory, tools, multiple agents, verification loops, or self-improvement infrastructure. A standardized harness ladder can therefore reveal model

properties that are invisible in single-turn benchmarks. It also lets researchers quantify whether a new harness architecture produces a general improvement across LLMs or only a model-specific effect.

21. Limitations and Open Questions

- The proposed level thresholds are hypotheses and require empirical calibration across domains and models.
- Task difficulty T is multidimensional; a scalar T band is useful operationally but should retain an underlying difficulty vector for horizon, uncertainty, ambiguity, verification burden, novelty, and environmental change.
- Cognitive Intelligence C may itself require a broad psychometric/domain profile rather than one benchmark family.
- Self-awareness is operational and evidence-grounded; the framework does not address phenomenal consciousness.
- O-levels are only meaningful for organizations with real state/authority/evidence boundaries, not a single prompt that role-plays several agents.
- Open-ended levels require long evaluation horizons and are especially vulnerable to benchmark contamination, resource confounds, and vague success criteria.
- Higher U denotes broader validated capability, not moral value, desirability, safety, speed, or economic efficiency.
- HLIS and HIL weights are provisional. They require calibration against external outcomes, sensitivity analysis, and anti-gaming studies; leaderboard use should publish the full component profile.
- A standardized harness ladder must control tools, context budget, compute, retrieval corpus, authority, and verifier access. Otherwise HIL becomes a measure of infrastructure spending rather than LLM ability.
- Benchmark-version drift is a first-class threat. HIL reports must freeze benchmark versions, hidden splits, harness code, tool inventories, and resource envelopes; otherwise apparent model progress may come from changing evaluations.
- HIL can inherit weaknesses from the benchmarks it integrates. A benchmark vector with contamination, narrow domain coverage, or weak verifiers does not become valid merely by being placed inside HIL.
- Harness Scaling Curves can be non-monotonic. A more complex harness may hurt a model through context overload, tool confusion, or protocol mismatch; HIL should report regressions rather than force monotonic scores.
- A composite HIL-Score should never be used without the benchmark vector and component profile. Different users may rationally prefer different points on the capability/autonomy/cost frontier.
- Long-term-memory evaluation is vulnerable to leakage and trivialization. If a benchmark leaves the complete history in the active context, it may measure long-context reading rather than persistence; if it measures recall only, it can reward stale or irrelevant storage. HIL memory suites therefore need true temporal discontinuities, update/contradiction cases, provenance checks, and downstream-usefulness measures.

- The pilot measured only Delegation Intelligence at H1 over a small HG0-HG3 ladder. It does not establish full HLIS or Unified Intelligence, and the upper HG4-HG Ω ladder remains uninstantiated in the reported pilot.
- The initial frontier task suite showed saturation for stronger executors; identical aggregate curves can therefore indicate insufficient benchmark headroom rather than model equivalence.
- Loss-weighted delegation utility depends on predeclared task-class cost ratios. The delivered-correct frontier, false-completion rate, false-rejection rate, human load, and raw outcome table should always remain available so application-specific utility assumptions can be changed without rerunning the model.
- Verification Responsiveness is sensitive to feedback quality and criterion granularity; comparisons require standardized, accurate failure messages.

22. Conclusion

A unified intelligence level is useful only if it summarizes rather than erases structure. This paper therefore proposes five conceptual intelligence families - Cognitive, Individual, Organizational, Delegation, and Self-Awareness Intelligence - while decomposing Delegation Intelligence into two directly measured coordinates, Task Difficulty and Human Cognitive Intervention, conditioned on verified reliability. The resulting scientific profile is [C, I, O, T, H, SA], while the theoretical presentation remains [C, I, O, DI, SA]. Long-term memory remains inside Individual Intelligence rather than becoming a sixth family: M is treated as the temporal substrate that enables I to progress from persistence to learning, governed self-improvement, recursive improvement, and discovery.

Unified Intelligence U0-U Ω is then defined as a cumulative gated hierarchy. Promotion requires retention of all lower-level capabilities and satisfaction of minimum C, I, O, SA and delegation-frontier requirements. This makes it legitimate to say that a higher U-level has a broader validated intelligence envelope, while avoiding the false claim that it must outperform every lower-level system on every narrow metric.

For harness-based AI, the framework provides both a development roadmap and a measurement program. Existing benchmark families remain visible as a capability vector; HIL adds the Harness Scaling Curve by freezing the LLM and progressively strengthening the harness. Long-term memory is embedded inside I rather than scored twice, so the curve can reveal whether the model actually exploits persistent episodic state, consolidation, verifier feedback, organizational structure, and self-improvement infrastructure. Version 1.5 also makes outcome integrity explicit: delivered-correct reliability defines the delegation frontier, false completion and false rejection are separately observable, and an optional loss-weighted net surface may enter continuous scoring. The broader methodological claim is that a harness-intelligence framework should be willing to revise its own measuring variables when building the ladder exposes a blind spot.

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Appendix A. Compact Reporting Template

Core measured profile: [C, I, O, T, H, SA] @ reliability p

Conceptual profile: [C, I, O, DI, SA], where $DI = F_A(H,p)$

Unified headline: $U_n [C_x, I_y, O_z, T_k/H_h, SA_s]$

Extended system profile: [U; C,I,O,DI,SA; λ ; D,A,R,V,S; resources, φ]

Field	Example
Unified headline	U3
Core profile	C5, I4, O3, T4/H1, SA4
Reliability	p = 0.80
Delegation frontier	$F_A(H0,.80)=T3$; $F_A(H1,.80)=T4$

Field	Example
Bottleneck	O3
Circle reach	$\lambda = (.42, .08, .03, 0, 0)$
Structural profile	D3, A2, V4, separated proposer/evaluator/promoter
Resource profile	Declared compute, time, memory, energy/cost budget
I-memory diagnostics (inside I)	Example: episodic/provenance 0.91; temporal update 0.84; consolidation/procedural transfer 0.76; not a separate HLIS axis
Outcome integrity	False-completion rate=0.00; false-rejection rate=0.01; VR=0.72 (example)

Appendix B. Suggested Unified Promotion Tests

Transition	Minimum new evidence in addition to full lower-level retention
U0 -> U1	Persistent cross-session state and episodic/task-memory recovery with provenance, temporal ordering, stale-state handling, and correct goal-level completion after restart/context discontinuity.
U1 -> U2	Verified experience is consolidated into durable semantic/procedural knowledge that improves held-out related tasks; knowledge updates/contradictions are handled; multi-step delegation under $\leq H2$.
U2 -> U3	Governed self/organizational method improvement plus durable failure/hypothesis/change/benchmark/version lineage, T3 strategy construction under $\leq H1$, and causal self-diagnosis.
U3 -> U4	Validated improvement of the improvement process using longitudinal meta-improvement history, plus long-horizon T4 work and accurate modeling of self-change.
U4 -> U5	Independent discovery of initially unknown solution method/knowledge with persistent hypothesis-experiment-evidence lineage and self-directed experiments about world and self-unknowns.
U5 -> U6	Mission -> projects -> tasks -> execution loop over changing conditions with persistent role/mission/decision history and grounded mission self-model.
U6 -> U Ω	Repeated bounded-charter mission generation whose validated discoveries create new tools/agents/organizations, with governed evolution of memory/cognitive instruments and preserved external lineage.

Framework status. This document is a theoretical proposal developed from the preceding Unified Intelligence, Delegation Intelligence, harness-indexed intelligence, and operational self-awareness design discussion. Level names and numerical thresholds are proposed working definitions and should be revised when benchmark evidence warrants.

Appendix C. Harness-Intelligence Reporting Template

Base LLM: model id/version/hash

Standardized harness ladder tested: HG0, HG1, ..., HGk

Per-harness: U* | HLIS | [C,I,O,T/H,SA] | delivered-correct reliability | bottleneck | false-completion/false-rejection rates | VR | resource budget | benchmark/ladder id

LLM summary: benchmark vector B(m) | Harness Scaling Curve HSC | HIL-Level | HLIS(HG0) | HIL-Ceiling | Harness Gain | VR | HIL-AUC/composite if reported with cautions | HDS if harness-design tasks were run

Application panel: selected coordinates such as I, O, SA, Circle lambda, verification strength, cost/latency - reported separately from the common core.

Appendix D. Version History

Version	Principal change
1.0	Five families, six measured coordinates, cumulative U0-U Ω hierarchy, HG0-HG Ω concept.
1.1	HLIS for a model-harness pair; HIL characterization across a standardized harness ladder.
1.2	Contemporary benchmark mapping and Harness Scaling Curve.
1.3	Formal HLIS construction, normalization, cumulative achievement, confidence/resource reporting.
1.4A	Long-term memory embedded inside Individual Intelligence; longitudinal I-Bench and memory diagnostics.
1.4B	Parallel measured-ladder draft: HG0-HG3 pilot, acceptance-invariance finding, false-completion-aware scoring, verification responsiveness, paired-rung protocol.
1.5	Merged revision: memory-integrated I plus outcome-complete delegation measurement, measured-ladder lessons, benchmark binding status, and curve-first HIL reporting.